

Pressure-Velocity Coupling in a Lid-Driven Cavity

SIMPLE, PISO and PIMPLE
A shared finite-volume Python implementation

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The central question

How does pressure enforce incompressibility, and how can different coupling loops reach the same flow with different computational costs?

Initial qualified case: $Re=100$. Two uniform grids. Executed fields, benchmark comparisons, momentum and continuity checks. An independent extension; Week 1 remains unchanged.

1. One physical problem, two formulations

Governing equations

$$\partial \mathbf{u} / \partial t + \operatorname{div}(\mathbf{u} \mathbf{u}^T) = -\operatorname{grad}(p) + \operatorname{Re}^{-1} \operatorname{Laplacian}(\mathbf{u})$$

$$\operatorname{div}(\mathbf{u}) = 0$$

Use a unit square, unit lid speed and density, and viscosity 0.01. All walls are impermeable; tangential velocities are zero except for the moving lid.

Week 1 solves streamfunction and vorticity: $u = \partial \psi / \partial y$, $v = -\partial \psi / \partial x$ and $\operatorname{Laplacian}(\psi) = -\omega$. Pressure is recovered separately from steady momentum. Week 1.2 solves velocity and pressure together.

Why add this extension?

Students can inspect pressure as the continuity constraint, shared-face mass budgets, under-relaxation, and physical-time versus coupling iterations. The pressure-velocity formulation also provides a route toward three-dimensional solvers.

Learning outcomes: derive the pressure operator; distinguish the three loops; verify mass and gauge invariance; compare measured errors and cost; state the limits of a steady benchmark.

2. Shared MAC finite-volume operators

Pressure p is stored at cell centers. Horizontal velocity u lies on vertical faces; vertical velocity v lies on horizontal faces. Arrays use y-first, x-second indexing.

Momentum and pressure algebra

$Aq = b - Gp$; $d = 1/\text{diag}(A)$
 $H(q) = b - \text{offdiag}(A)q$; $q = d(H - Gp)$
 $L = -D d G$; $Lp = -D(dH)$

D computes the net velocity flux divided by cell volume. A shared face contributes with opposite signs to its two neighbors. This gives exact interior flux cancellation.

Normal wall velocities are zero. Tangential wall diffusion uses half-cell distances. The lid-corner discontinuity is retained. Central convection and diffusion use the same frozen advecting face velocities for every coupling method.

Gauge and compatibility

L has a constant nullspace. The closed-cavity RHS must sum to zero. Pin one cell during the sparse solve; shift reported pressure to zero mean. A pressure constant cannot change any velocity correction.

Sparse LU is a linear solver, independent of the pressure-coupling algorithm. No multigrid or packaged CFD solver is used.

3. SIMPLE: steady correction and relaxation

1. Build and under-relax steady momentum using the current velocities.
2. Predict face velocities from the current pressure.
3. Solve $Lp' = -Dq^*$.
4. Correct velocity fully: $q = q^* - dGp'$.
5. Update pressure $p \leftarrow p + \alpha_p p'$; repeat until momentum and continuity pass.

Recorded settings

$\alpha_u = 0.7$; $\alpha_p = 0.3$. Steady momentum $Linf < 10^{-7}$; continuity uses the same declared tolerance.

Advantage: transparent steady pressure coupling and useful relaxation experiments.

Limitation: sensitivity to relaxation; iteration count is not physical time or a fair cost measure.

Worked mass correction

If a tentative field has a nonzero cell divergence, pressure correction changes shared face fluxes until the corrected net flux vanishes. Independent velocity clipping would generally destroy that conservation property.

4. PISO: corrections inside a physical step

1. Keep old-time velocities fixed.
2. Assemble and solve backward-Euler momentum for tentative velocity.
3. Recompute H from the current velocity; solve $Lp = -D(dH)$.
4. Correct $q = d(H - Gp)$.
5. Re-evaluate off-diagonal momentum contributions and repeat the pressure/velocity correction.

Recorded settings

$\Delta t = 0.05$; two pressure corrections per physical step. The old-time source remains fixed within both corrections.

Advantage: additional pressure-momentum consistency without a complete nonlinear outer loop.

Limitation: more pressure solves per step; temporal accuracy still depends on Δt and the time discretization.

Do not mislabel a solver

Two Jacobi/LU iterations on a single pressure system are not two PISO corrections. The extra correction must update the momentum coupling using the corrected velocity.

Backward Euler has first-order temporal accuracy. Marching to steady state does not validate startup trajectories.

5. PIMPLE: outer momentum iteration plus PISO

Inside one physical time step, repeat momentum assembly and prediction, then the PISO correction loop. The new velocity changes nonlinear convection coefficients for the next outer loop.

Recorded settings

Three outer loops; two pressure corrections in each; $\Delta t=0.05$. Old-time velocities remain immutable across all six pressure corrections.

Advantage: additional nonlinear consistency within each time step; useful when a single PISO pass is insufficient.

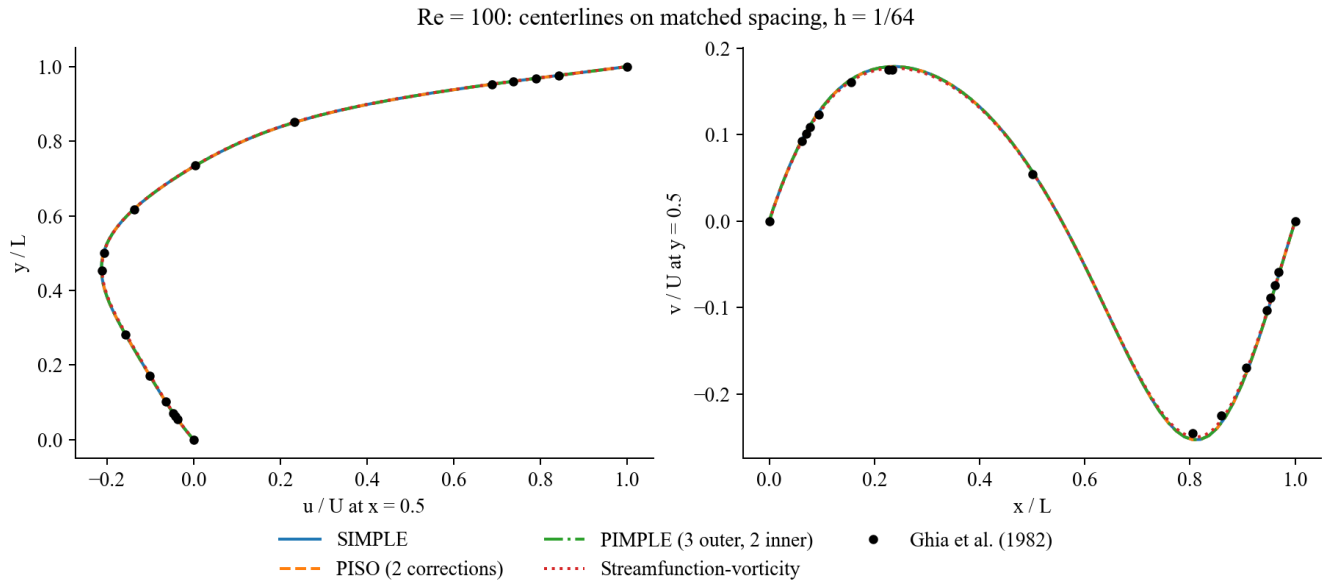
Limitation: repeated momentum and pressure solves increase cost. Larger stable steps do not guarantee accurate transient dynamics.

An exact implementation check

Set the number of outer loops to one. With identical settings, the resulting PIMPLE step must match this PISO step exactly.

For this simple steady $Re=100$ cavity, extra outer loops need not be economical. A time-dependent lid and a matched temporal-error target are better tests of potential PIMPLE benefits.

6. Benchmark velocity and two-grid evidence



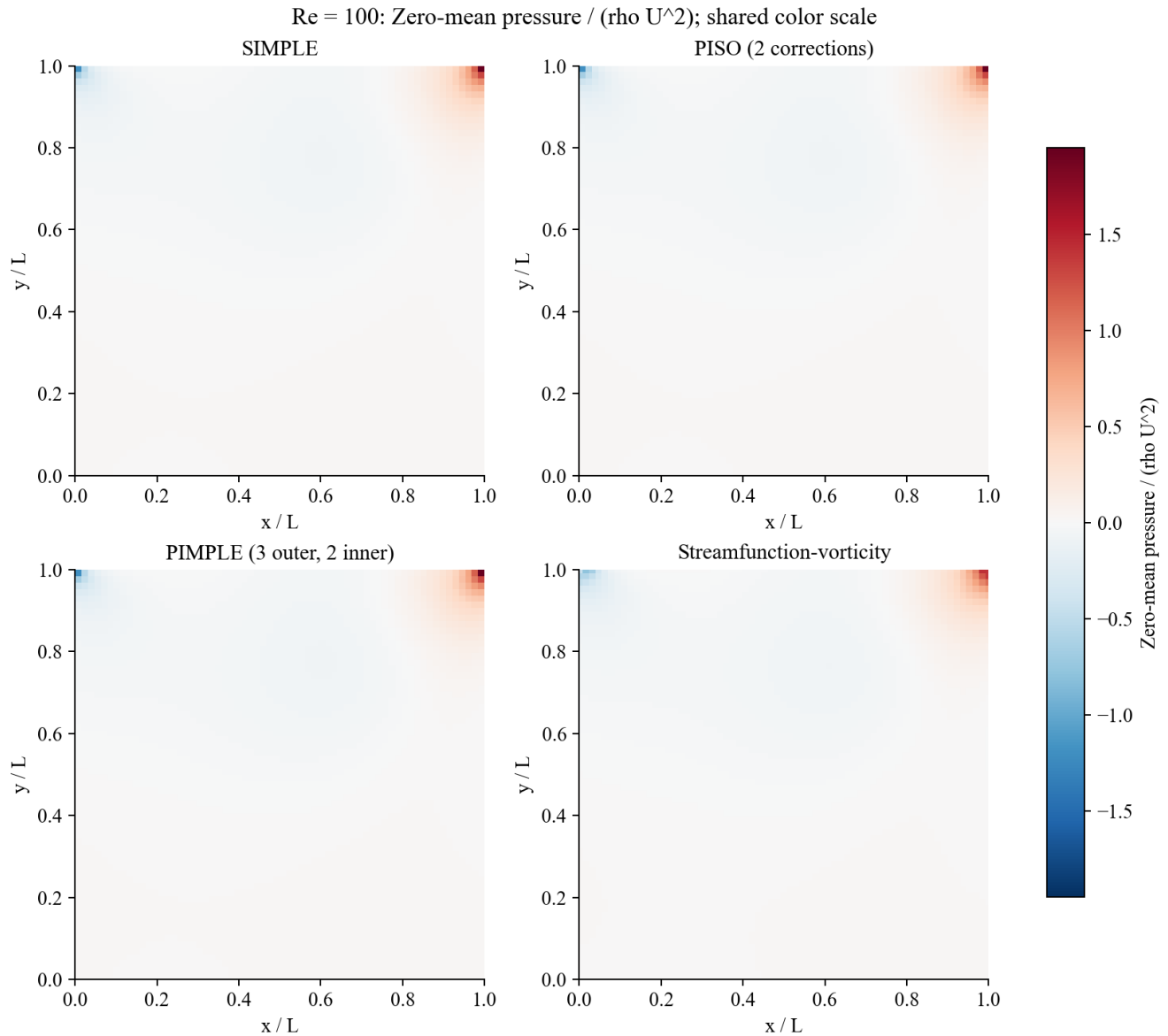
Ghia, Ghia and Shin (1982) velocity markers are compared at their stated coordinates, including wall endpoints. MAC face locations and FD nodes are interpolated only for the declared centerline diagnostics.

Fair resolution comparison

32/64 FV cells per side correspond to 33/65 FD nodes: compare h , not array length. The three FV algorithms use identical spatial equations. The independent FD formulation is a cross-check with a different truncation error.

The FD reference errors decrease on refinement; FV errors against Ghia increase slightly between these two grids. This nonmonotone trend is retained explicitly. Coarse-grid agreement with discrete benchmark data does not establish continuum accuracy. Further refinement and another reference are needed; no asymptotic order is claimed. Ghia velocity tables do not validate pressure.

7. Physical fields and pressure interpretation



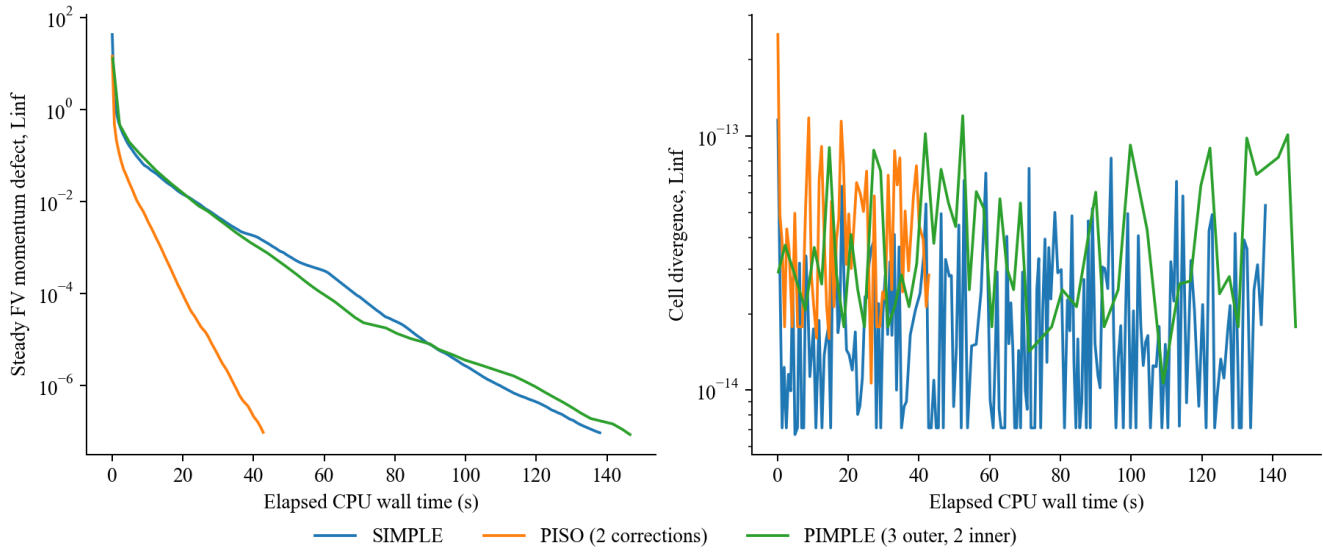
All pressure panels use the same zero-mean convention and common color limits. The three converged FV fields agree closely; the FD recovered pressure can differ near the singular lid corners.

Pressure evidence

Comparing pressure between independent formulations is informative but is not an exact-pressure validation. The notebook also shows speed, streamlines and vorticity on shared physical scales.

8. Convergence, cost and stopping conditions

Coupling convergence: one measured local CPU solve per method



The plotted steady FV momentum defect has units U^2/L , with $U=L=1$. Continuity is the maximum cell divergence. The elapsed-time axis includes actual numerical work, so differences in loop cost remain visible.

The FD reference uses a vorticity-update stopping norm and is not overlaid as if it were the same FV residual. A small under-relaxed update is not a substitute for a momentum-equation residual.

Measured conclusion

All six new FV runs reach the declared steady tolerance. On each grid, the three methods agree within 2×10^{-6} in u , v and zero-mean p . Timings are one local CPU solve per method, not a universal speed ranking.

9. Exercises and the next experiment

1. Sum cell divergence and cancel shared interior face terms.
2. Add a pressure constant; verify unchanged velocity.
3. Compare one and two PISO corrections using the one-step momentum defect.
4. Check PIMPLE with one outer loop against PISO.
5. Change SIMPLE relaxation at fixed final residual tolerance.
6. Change only convection interpolation; separate spatial error from coupling effects.
7. Design a Δt -halving startup experiment before claiming a PIMPLE speedup.

Next qualification

Start with a ramped or oscillating lid. Use the same temporal scheme and matched final times. Refine Δt ; measure field and phase errors. Any transient streamfunction pressure reconstruction must include acceleration terms.

Only $Re=100$ steady behavior is qualified in this release. Higher- Re behavior, transient accuracy and stretched grids remain separate tasks.

10. Reading, attribution and reproducibility

PySIMPLE: educational MAC finite-volume SIMPLE

NIST FiPy: pressure-correction derivation (Stokes example)

OpenFOAM: solution and algorithm control

pyOpenFOAM: another inspectable Python loop organization

Ghia, Ghia and Shin (1982): benchmark tables

This independently written module uses original implementation and explanations. External examples inform derivation and organization; their validation claims are not inherited. The audited CFD-Python PISO-titled notebook had a commented-out second correction, illustrating why source inspection matters.

Reproduce and audit

qa/run_week01_2.py executes all cases; qa/build_week01_2.py builds figures and the notebook.
tests/test_pressure_velocity.py checks flux cancellation, pressure nullspace, PISO correction behavior, PIMPLE limits and the common steady solution. Results retain source and field hashes and runtime versions.

Lecture and code are English. Typography: Times New Roman. Prepared 09 October 2026.